makes a story more similar to the average of other stories within the same condition (*Human with 1 GenAI idea b*=0.871, *p*<0.001; *Human with 5 GenAI ideas b*=0.718, *p*=0.003).

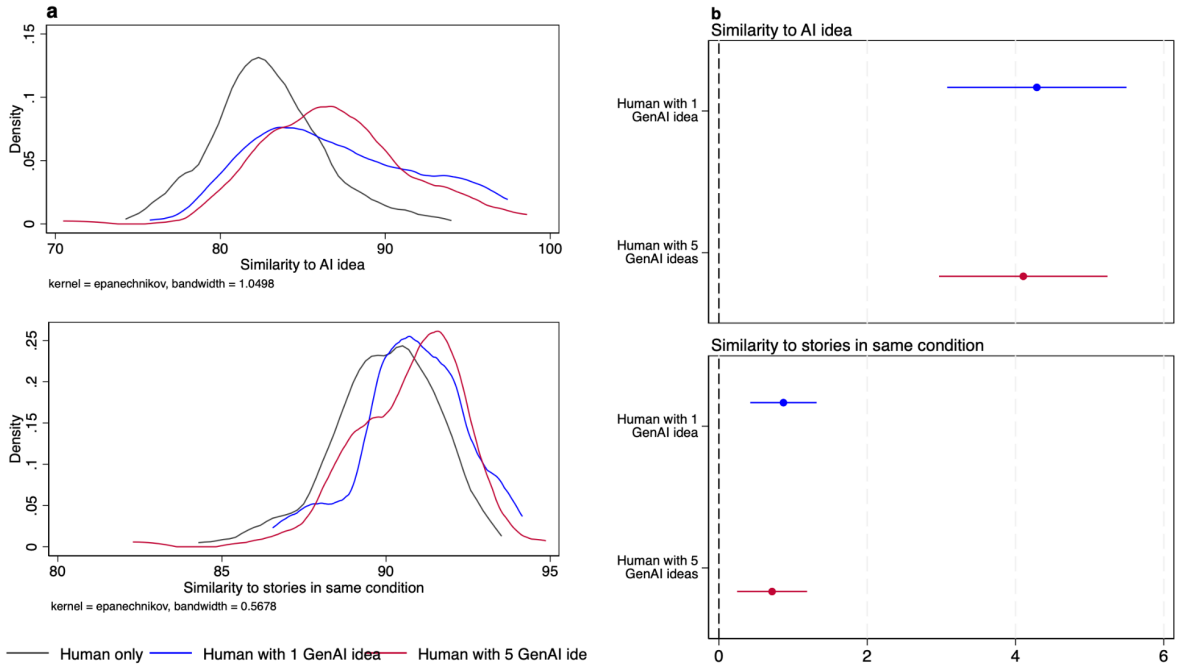

Figure 5. Comparison of similarity of writer stories to GenAI ideas and others stories. **a**, Kernel density plots comparing story similarity to ideas produced by GenAI and all other stories in the same condition for each condition. **b**, Compares story outcomes of *Human only* (reference category) to humans with access to 1 and 5 GenAI ideas.

Discussion

GenAI has the potential to dramatically impact most aspects of the economy and society at large (*21,22*). Previous empirical work has focused on its effects on productivity, routine tasks, sales, resume writing, and joint collaboration between humans and AI, including for scientific and medical tasks (*3–6,23–25*), all of which contribute to our understanding of the potentially transformative impact of GenAI. In this paper, we extend this work by studying a question fundamental to all human behavior, which is of both economic and purely expressive value: how does generative AI affect human creativity?

We offer an answer to this question by experimentally studying the causal effect of having access to GenAI on writing short stories in an online experiment. We find that having access to GenAI causally increases the average novelty and usefulness—two frequently studied dimensions of creativity—relative to human writers on their own. This is driven, in particular, by our experimental

condition that enables writers to request multiple GenAI ideas—up to five in our study—each presenting a different starting point, leading to a "tree" branching off to potential storylines (*3*).

Our results provide insight into how GenAI enhances creativity. Having access to GenAI "professionalizes" the stories beyond what writers might have otherwise accomplished alone. The overall effect is a more novel, more useful story that is well written and enjoyable. However, the gains from writing more creative stories benefit some more than others: less creative writers experience greater uplifts for their stories, seeing increases of 10% to 11% for creativity and of 22% to 26% for how enjoyable and well written the story is.

We note two additional observations about these results. First, having access to GenAI effectively equalizes the evaluations of stories, removing any disadvantage or advantage based on the writers' inherent creativity (*26*). That GenAI particularly benefited less able writers is paralleled in recent studies focusing on other domains in which GenAI has been shown to help less productive workers (*4,5*). Second, one might ask whether the GenAI ideas can push the upper bound of creativity of produced stories, beyond what particularly creative humans are capable of on their own. We do not find evidence of this possibility in this study; however, to test this hypothesis more thoroughly, we encourage future research to allow creatives to engage with GenAI more deeply and interactively, and to extend this work by leveraging more specialized GenAI tools for this purpose.

Our choice of the experimental design offers a fairly stringent test to measure the causal impact of GenAI on creativity. We designed our study such that endogenous decisions by the writer are minimized, but not fully eliminated. We do not allow writers to customize the call to the GenAI engine, nor do we allow for repeated interactions between writers and GenAI, both of which may increase the effectiveness and magnitude of the impact of GenAI on creativity. If that is the case, our estimates are likely a lower bound of the potential that GenAI could offer to writers when they are given full control over the AI engine, or when real-time interactions are enabled that help writers with ideation and enhancement further (*27*). Indeed, previous research suggests that conversations with others can improve ideation (*28*)—a possibility that GenAI could facilitate and enable in new and faster ways than ever before. That a tightly-controlled prompt requesting a GenAI idea shows significant effects on creativity in our study provides a promising starting point for future researchers to delve deeper into customization and personalization of GenAI for different writers (*8*).

We do, however, allow writers to opt into receiving GenAI ideas, rather than assign GenAI ideas to everyone in the GenAI conditions. We do this to ensure that writers are invested in, and receptive to, what GenAI produces. Furthermore, we anticipated that—if offered—the vast majority of participants would take advantage of the option to at least see the GenAI idea, thus minimizing the risk of self-selection affecting our causal estimates. The empirical evidence shows that nearly 9 out of 10 people in the GenAI conditions choose to receive at least one GenAI when offered, bolstering our confidence that our results—based on our conservative intention-to-treat analysis that studies the effect of condition regardless of whether writers did or did not choose to request GenAI ideas—allow for a causal interpretation.

Much has been written about the potential replacement of AI on human labor (e.g. automation) (*29,30*) or a "horse race" between human and AI-generated ideas (*31*). We focus on the potential complementarities of AI on human creative production. We do so among a sample of relatively "typical" study participants often used in academic studies (which comes with limitations on population representativeness) (*32*)—that is, we do not study professional writers or unusually creative individuals. These individuals remain an important but understudied population segment, for which the effects of GenAI could be transformative in other ways, potentially offering efficiency gains or improved speed of execution (*6*). That said, our results suggest that GenAI may have the largest impact on individuals who are less creative.

Despite the enhancement effect that GenAI had on average creativity, we also believe our study offers a cautionary note if GenAI were adopted more widely for creative tasks. In general equilibrium, an interesting question is whether the stories enhanced and inspired by AI will be able to create sufficient variation in the outputs they lead to. Specifically, if the publishing (and self-publishing) industry were to embrace more GenAI-inspired stories, our findings suggest that the produced stories would become less unique in aggregate and more similar to each other—an effect that may be heightened over time, as many GenAI models use the existing corpus of text and published literature to train and refine their algorithms and predictions, thus further reducing diversity. Initial evidence suggests that GenAI models trained with GenAI content become unstable (*33*).

GenAI is a rapidly evolving technology with its full potential yet to be explored. While our study used the most recent version of a widely used GenAI tool—OpenAI's GPT-4—this may soon become obsolete. However, rather than limiting our study or future studies, we believe the fast progress of GenAI development offers exciting opportunities for researchers interested in creativity, art, literature, and innovation. If GenAI leads to enhancements of human creativity in a conservatively designed experimental study today, the creative possibilities for tomorrow may extend beyond our current, collective imagination.

Methods

Writer study and experimental conditions. For the Writer Study, we recruited 500 participants to participate in the experiment from the Prolific platform. Using the platform's filtering options, we included participants who were Prolific participants who indicated that they are based in the United Kingdom with an approval rating of at least 95% from between 100 and 1,000,000 prior submissions. Writers are not selected based on prior writing skills or their creativity. Of the 500 participants who began the study, 169 exited the study prior to giving consent, 22 were dropped for not giving consent, and 13 dropped out prior to completing the study. Three participants in the *Human only* condition admitted to using GenAI during their story writing exercise and—as per our pre-registration—they were therefore dropped from the analysis, resulting in a total number of writers and stories of 293.